

Indian Banking Credit Dashboard

Project Narrative & Analysis Report

Economic Logic • STAR Framework • Challenges & Solutions • Key Conclusions

Data Source: Reserve Bank of India — DBIE / Handbook of Statistics on the Indian Economy
Analysis Period: FY 1996–97 to FY 2025–26

18 Datasets | 5 Dashboard Pages | 20 Charts | 36 States/UTs

1. Why This Project Exists — and Why It Matters

The Problem with Banking Data in India

India's banking system is one of the most consequential institutions in the economy — it determines who gets credit, at what cost, and in what quantities. Yet the data describing it is scattered across dozens of separate RBI publications, formatted inconsistently, split across multiple time periods, and almost never presented in a form that allows someone to ask a genuinely cross-cutting question: does credit growth track GDP? Do states with high CD ratios also have worse NPAs? Has the digital payments revolution changed deposit behavior?

This project exists to answer those questions from first principles, using only publicly available RBI data — no paid terminals, no Bloomberg subscriptions, no proprietary datasets. Every number in this dashboard can be traced back to a specific RBI publication table.

Who Needs This — and Why Now

- Banking analysts who want to contextualize individual bank performance against the system backdrop.
- Policy researchers studying the effectiveness of RBI mandates like Priority Sector Lending — where a 40% ANBC target exists but compliance is rarely visualized.
- Students and academics building an understanding of how monetary policy cycles (repo rate) transmit into credit quality (NPA) and deposit behavior (CASA).
- Anyone trying to understand India's financial inclusion story — SHG linkage, agricultural credit growth, and UPI adoption are all parts of the same structural shift.

2. Economic Logic Behind Each Pillar

Sectoral & Industry Credit Deployment

Bank credit is the primary transmission mechanism from monetary policy to the real economy. Where credit flows determines which sectors expand — a sustained shift toward Personal Loans and Services over Industry signals a consumption-led growth phase rather than a private capex cycle. Tracking this prevents misreading GDP growth as investment-driven when it is actually consumption-driven, which has very different implications for long-run productivity and vulnerability to interest rate shocks.

Priority Sector Lending

Credit markets fail in specific, predictable ways — they under-serve agriculture, small enterprises, and affordable housing because these borrowers lack collateral, have high monitoring costs, and generate returns that don't price in social value. RBI's 40% ANBC mandate exists precisely to correct this market failure. Tracking priority sector disbursements (especially the Housing and Social Infrastructure acceleration visible in this data) shows whether policy intent translates into actual capital allocation — or whether banks are gaming sub-targets.

CASA Ratio (Deposit Mix)

A bank's CASA ratio directly determines its cost of funds. Current and savings deposits are essentially zero- or near-zero-cost liabilities; term deposits are priced competitively against other instruments. A bank with 45% CASA can lend 200-300 basis points cheaper than one with 25% CASA while maintaining the same net interest margin — which is the fundamental competitive advantage of deposit-rich retail banks. System-wide CASA also reflects household liquidity preference: rising CASA during uncertainty (more transactional savings), falling CASA during rate tightening cycles (shift to term deposits for yield).

Non-Performing Assets (NPA)

NPAs are the single most important lagging indicator of credit cycle health. When NPA ratios rise, banks face capital erosion, provisioning pressure, and a rational incentive to pull back on new lending — creating a credit crunch that feeds back into economic slowdown. The 2015-2018 Asset Quality Review forced Indian banks to recognize hidden NPAs, causing GNPA to spike to 11.2% — the mechanism that explains both the investment drought of 2016-2019 and the credit growth slowdown that followed. The subsequent recovery to 2.75% (FY2023-24) marks the single most significant balance sheet healing event in Indian banking history in two decades.

Digital Payments (UPI / IMPS / NEFT)

Digital payments growth is not just a technology story — it is an infrastructure story with second-order credit effects. A UPI-linked bank account creates a transaction data trail that enables underwriting of previously thin-file borrowers, making credit inclusion economically viable for the first time. The ~10,000x growth in UPI volume from FY2016-17 to FY2024-25 represents a structural shift in India's financial infrastructure comparable to the introduction of RTGS in the 2000s — except this time, the beneficiaries are retail and small-business borrowers rather than large corporate treasuries.

State-wise Credit-Deposit Ratio

The CD ratio tells you whether a state is a net credit-deployer or a net saver in the banking system. States like Andhra Pradesh (155%) and Tamil Nadu (120%) extend more credit than they collect in deposits — funding the gap through inter-bank borrowing, which implies their local economies depend structurally on credit intermediated from surplus-deposit states. States like Uttarakhand (43%) and Himachal Pradesh (39%) are net savers — their depositors are effectively funding credit deployment elsewhere. This geographic dimension is entirely invisible in national aggregate data and has direct implications for regional policy design.

Macro Overlay (GDP, Repo Rate, CPI)

Banking metrics do not exist in a vacuum. The NPA crisis of 2015-2018 is incomprehensible without seeing that it coincided with elevated real interest rates (repo above CPI) and decelerating GDP growth — a classic credit cycle contraction. The COVID dip in NPA recognition (2019-2021) similarly requires the GDP collapse context to understand why it was a recognition artifact rather than genuine improvement. Overlaying these macro series transforms a banking dashboard from a descriptive report into an analytical tool.

3. STAR Framework — Situation, Task, Action, Result

S — SITUATION

India has 18+ Scheduled Commercial Bank data series published by RBI across separate databases (DBIE), bulletin tables, quarterly publications, and annual handbooks. Each uses different time periods, different base years, inconsistent fiscal year labeling (some use April-March FY strings like '2023-24', others use calendar years), and multi-row merged headers that break standard CSV parsers. No single consolidated, analysis-ready dataset existed covering credit, deposits, asset quality, digital adoption, and macro context simultaneously. Analysts wanting a cross-pillar view were forced to manually stitch together data from at least 6-8 different downloads per analysis session.

T — TASK

Build a complete, reproducible, analysis-ready Indian banking dashboard covering seven pillars — sectoral credit, priority sector lending, CASA, NPAs, digital payments, financial inclusion (SHG), and state-wise credit-deposit ratios — overlaid with macroeconomic context (GDP, inflation, repo rate). The deliverable had to be a real, usable dashboard (Tableau-ready cleaned CSVs + TWBX workbook), not just a description of what such a dashboard would contain. Every number had to be traceable to a specific RBI source table, with all data quality issues surfaced and documented rather than silently corrected.

A — ACTION

Seven distinct actions were taken across the project lifecycle:

1. Source identification — navigated the DBIE publication tree across 5+ categories to identify the precise 18 tables covering all required pillars, rejecting ~40 candidate tables that were discontinued, duplicated, or covered structurally different institution types (co-operatives, RRBs) that would corrupt system-level ratios.
2. Format analysis — inspected each raw XLSX for multi-row headers, merged cells, forward-fill patterns, wide-vs-long format differences, and fiscal year labeling inconsistencies before writing any cleaning code.
3. Systematic cleaning — wrote Python/pandas cleaning scripts for all 18 files, handling: (a) merged-cell resolution via `openpyxl merge_map` construction, (b) header forward-filling for wide DBIE export formats, (c) wide-to-long reshaping for bank-group and state-level data, (d) fiscal year standardization to a common 'YYYY-YY' key across all sources.
4. Cross-pillar unification — merged NPA, UPI, GDP, agri credit, and CASA onto a common FY dimension to enable correlation analysis. Surfaced that NPA and UPI volume have $r = -0.94$ correlation (2016-2024), a coincident finding driven by the same post-2018 banking clean-up cycle.
5. Forecasting — fitted floor-adjusted linear trend models for NPA (structural floor ~1.8%) and CASA ratio (+0.67 ppt/year), with explicit caveats that these are indicative projections, not econometric models.
6. Visualisation — rendered 20 charts across 5 dashboard pages, verifying each visually and correcting 3 bugs caught mid-render (wrong industry filter, year mismatch, unsorted categorical x-axis).
7. Documentation — built a full GitHub README with economic logic, findings, limitations, and an automation script template for refresh pipelines.

R — RESULT

Final deliverables: 18 clean CSVs totaling ~200KB of structured data spanning FY1950-51 to FY2025-26; a Tableau TWBX workbook with 18 connected data sources, 18 pre-built worksheets, and 5 dashboard tabs; a 5-page PDF dashboard with 20 verified charts; and full project documentation.

The dashboard closed 6 of 7 originally planned pillars — the only gap being state-wise credit data, which was eventually sourced from the Quarterly Spatial Distribution publication and is now included.

Six specific analytical findings emerged from the data that were not known at the project start — these are documented in Section 5 below.

4. Challenges Faced and How They Were Tackled

Challenge 1: Multi-row merged headers in DBIE exports

How it was tackled: RBI's XLSX exports use 3-6 merged header rows, with meaningful column labels spread across rows 6, 7, and 8 depending on the table. Standard `pd.read_excel()` treats all but the first as data, producing garbage column names. Tackled by: reading with `openpyxl` directly, building a `merge_map` that forward-fills merged cell values across their full span, then reconstructing clean (quarter, metric) tuples before reading any data rows.

Challenge 2: Fiscal year label inconsistency across sources

How it was tackled: DBIE tables use at least four different year representations: '2023-24', '2023-2024', 2024 (integer), and 'Mar-2024' (month-year). Joining these on a common key without standardisation produces silent wrong matches (e.g., 2024 matching '2023-24' or '2024-25' depending on rounding). Tackled by: writing a fiscal year normaliser function applied to all sources before any merge, mapping every format to 'YYYY-YY' strings.

Challenge 3: Uppercase state names filtering incorrectly

How it was tackled: An early version of the state-wise extraction code filtered out rows where state names were all-uppercase, assuming these were subtotal rows. In RBI's formatting, ALL state names are uppercase (CHHATTISGARH, ANDHRA PRADESH) while subtotal rows have None in the state column. The filter removed all valid state data and returned zero rows. Caught by printing raw row values before any filtering; fixed by filtering on None rather than case.

Challenge 4: DBIE server load errors during data collection

How it was tackled: The DBIE server returned 'reports currently unavailable' errors during peak business hours (approx 12-3pm IST), blocking download of the Quarterly Spatial Distribution file. Tackled by: attempting downloads during off-peak hours and keeping a clear list of exactly which files were still needed so re-attempts were targeted, not exploratory.

Challenge 5: Wide-format DBIE exports (years as columns)

How it was tackled: Several DBIE tables store time series wide — each year is a column, each row is a variable. This is incompatible with both Tableau's data model (which expects one row per observation) and pandas groupby operations. Tackled by: identifying which tables needed wide-to-long reshaping before building cleaning scripts, and using `pd.melt()` or manual long-format construction for those tables.

Challenge 6: Stale deposit-side state data

How it was tackled: The 'Bank Deposits of SCBs - District Wise' annual series stops at 2018 while the credit-side equivalent continues to 2026. Using both would create a silent 8-year mismatch in any CD ratio calculation. Tackled by: sourcing the Quarterly Spatial Distribution file (Statement 3A) instead, which has both deposits and credit in a single consistent table through Q4 FY2025-26, eliminating the staleness problem.

Challenge 7: Structural floor in NPA forecast

How it was tackled: A naive linear extrapolation of the FY2018-2024 NPA decline produced a forecast of 0.23% by FY2025-26 — an unrealistic number since no major banking system sustains below 1.5-2% GNPA. Tackled by: capping the forecast at a structural floor of ~1.8% based on long-run international banking minima, and explicitly labelling the forecast as indicative rather than model-based in both the chart and documentation.

Challenge 8: Correlation vs. causation (UPI vs NPA)

How it was tackled: The $r = -0.94$ correlation between UPI volume and NPA ratio sounds striking but is almost certainly a coincident correlation — both trends run from 2016 to 2024, driven by the same post-AQR banking environment and digital push policy, not a causal link from UPI to NPA reduction. Tackled by: including an explicit caveat annotation on the scatter chart and in the documentation, preventing the finding from being misrepresented.

5. Key Conclusions — What the Data Actually Shows

These are findings that emerged directly from the cleaned data — numbers and relationships that were not known at the start of the project and cannot be found in any single RBI publication as stated.

01

India's NPA crisis has fully reversed — and the recovery is near-historic

System-wide GNPA peaked at 11.2% in FY2017-18 following the Asset Quality Review and fell to 2.75% by FY2023-24 — better than every year between FY2011 and FY2021, approaching the pre-2008 lows. Public Sector Banks remain higher (3.5%) than Private (1.8%) and Foreign (1.2%), but the gap has narrowed sharply. This is the most significant balance sheet healing event in Indian banking in at least two decades.

02

Services and Personal Loans have displaced Industry as the engine of credit growth

Services sector credit grew 18.6% YoY and Personal Loans grew 16.0% — both ahead of Industry (15.1%) and Agriculture (13.7%). This is a structural, not cyclical, shift: for the first time since bank nationalisation, retail/services lending is the primary driver of system credit growth. This has implications for NPA risk (personal loan books are more vulnerable to income shocks) and for the traditional banking model of relationship-based corporate lending.

03

UPI has grown ~10,000x in transaction volume since launch — and still accelerating

From ~1.8 crore transactions in FY2016-17 to ~18,587 crore in FY2024-25. Value grew from Rs.6,961 crore to Rs.26 lakh crore over the same period. RTGS still dominates by value (Rs.2,013 lakh crore in FY2024-25) because it carries large corporate settlements — but UPI has displaced both NEFT and IMPS as the retail payment rail of choice, and the volume curve shows no sign of plateauing.

04

Six Indian states have CD ratios above 90% — a geographic credit concentration risk

Andhra Pradesh (155%), Tamil Nadu (120%), Telangana (112%), Maharashtra (100%), Rajasthan (93.5%), and NCT of Delhi (88.6%) all have very high CD ratios. The top 3 states have more credit outstanding than deposits collected locally — funding the gap through inter-bank. Meanwhile, 12 states have CD ratios below 50%, meaning more than half their deposits are intermediated out of their local economy into high-CD states.

05

Priority sector lending shows a significant internal divergence

Housing (+39.9% YoY) and Social Infrastructure (+35.3%) are the fastest-growing priority categories. Export Credit (-12.0%) and 'Others' (-13.5%) are the only contracting categories. This suggests either deliberate de-risking of export-linked books (consistent with global trade uncertainty) or RBI target reclassification effects. The aggregate priority sector figure (used for ANBC compliance) masks this divergence entirely.

06

The NPA crisis and the digital payments surge are coincident, not causal

The $r = -0.94$ correlation between UPI volume and GNPA from FY2016-17 to FY2023-24 is statistically striking but almost certainly reflects shared causation — both trends were driven by the post-2018 environment of aggressive provisioning, banking clean-up, and a government-led digital push. Claiming UPI caused NPA improvement would be methodologically unsound. The correct reading is that both happened simultaneously in the same policy environment, and disentangling the causal channels requires loan-level microdata.

6. Limitations — What This Dashboard Cannot Tell You

■ National aggregates hide bank-level and loan-level variation

Every system-wide metric (CASA ratio, GNPA, credit growth) is an aggregate that smooths over enormous variation at the individual bank and loan level. A system CASA of 13% is consistent with one bank at 45% and another at 5%. A system GNPA of 2.75% is consistent with some bank-industry combinations at 15% while others are near-zero. Identifying stress before it becomes systemic requires loan-level or bank-level data that this project does not use.

■ CASA ratio is system-wide, not bank-comparable

The CASA ratio computed from the Select Aggregates fortnightly data is a weighted average across all scheduled commercial banks. It cannot be compared to any individual bank's published CASA ratio without explicit caveat, and even then the comparison is misleading because bank-level CASA includes savings accounts separately from current accounts, which this aggregate does not always distinguish.

■ Forecasts are trend projections, not econometric models

The NPA and CASA forecasts use simple linear trend extrapolation on recent data. They do not model macroeconomic shocks, credit cycle turning points, RBI policy changes, or sectoral stress events. They are directional indicators only and should not be used for any investment or policy decision.

■ Time-period inconsistency across pillars

Different datasets have different historical depths: NPA data goes to 1996, CASA data to 2024, UPI data to 2015-16, state CD ratio data to 2022-23. The unified annual table is therefore sparse outside the 2016-2024 overlap window. Cross-pillar correlations computed on only 8 years of overlap should be treated as indicative, not conclusive.

■ Priority sector compliance cannot be verified from this data alone

This dashboard shows priority sector outstanding amounts but not the ANBC denominator required to compute compliance ratios. A bank reporting high priority sector disbursements may still be non-compliant if its ANBC grew faster. Full compliance analysis requires bank-level ANBC data from STRBI tables not included in this version.

■ No stress-testing or scenario analysis

The dashboard is retrospective and descriptive. It cannot answer: 'what happens to NPA if GDP grows at only 5% for two years?' or 'how would a 100bps repo rate hike affect CASA ratios?' Answering those questions requires a structural model of the banking system, not a data aggregation exercise.

All data sourced from the Reserve Bank of India — Database on Indian Economy (DBIE) and Handbook of Statistics on the Indian Economy. All underlying data is in the public domain. This analysis and the cleaned datasets are independently produced derivative works.